

When Does the Body Matter? A Dynamical Signature of Coupling Dependence in Evolved Neural Controllers

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Abstract. The debate over whether the body is constitutive of cognition or merely causally necessary for it has not yielded to behavioral evidence: any pattern observed in the behavior of an embodied agent is compatible with both views. This paper presents a method for studying the role of embodiment that focuses on neural dynamics data instead of behavior data. It asks the question, under what conditions do a controller’s neural dynamics depend on continuing sensorimotor coupling, and why? Existing ghost-condition methods, developed for behavioral metrics, fail when the dependent variable is neural-state divergence; sensory replay to a deterministic network yields zero divergence by construction. The paper proposes three corrected interventions that vary input rather than replay it, applies them across 60 phototaxis-evolved CTRNNs, and characterizes the dynamical regime that emerges. It is demonstrated that networks with high perturbation growth rate, input-sensitive attractor landscapes, and high inter-trial trajectory variance ($\rho = 0.80$, the strongest single predictor) exhibit high embodiment dependence. Network size shifts the probability that evolution discovers this regime but does not determine it. The signature aligns with the edge-of-chaos regime in reservoir computing and with the input-driven attractor structures of trained recurrent networks, suggesting that embodiment dependence may be a general property of recurrent systems that compute on streaming input.

Keywords: embodied cognition · ghost conditions · CTRNN · dynamical systems · embodiment dependence · evolutionary robotics · edge of chaos

1 Introduction

The embodied cognition debate over whether the body is constitutive of cognition or merely causally necessary has not been settled by behavioral evidence. I take this as suggestive that embodied cognitive researchers ought to look for new sources of evidence. The problem is that behavioral dependence on sensorimotor coupling is consistent with both interpretations; what distinguishes them is a metaphysical commitment about how to carve the cognitive system, not a

pattern that any single behavioral measurement could pry apart. In what follows I bracket that metaphysical question and ask a tractable empirical one: *under what conditions do neural dynamics become dependent on ongoing sensorimotor coupling, and why?*

I term this property *embodiment dependence* (ED): the degree to which a system’s neural trajectory diverges from its embodied operating regime when the sensorimotor loop is interrupted. ED is conceptually distinct from constitutive embodiment. A network can depend heavily on coupling without the body being a proper part of the cognitive mechanism in any sense Adams and Aizawa [1] would resist, and a network can be constitutively embodied in the sense articulated by Di Paolo et al. [12] while remaining largely independent of moment-to-moment input. The reverse is also possible: the brain-in-a-vat scenario illustrates one direction of dissociation. ED is operationalized through interventions on the loop rather than through reasoning about boundaries, in the tradition that runs through Pfeifer and Bongard’s analysis of how morphology shapes cognition [18] to the dynamical-systems work on minimally cognitive agents [3, 6].

This paper makes three contributions. First, I propose corrected ghost conditions that make neural-state divergence a usable interventional measure of embodiment dependence. The methodological extension complements the behavioral-divergence ghost-condition tradition [11, 4, 15], which has used these conditions productively for behavioral metrics where the choice of dependent variable does not introduce the deterministic-update constraint discussed in Section 2. Second, I identify a dynamical signature of the coupling-dependent regime: networks with high perturbation growth rate, input-sensitive attractor landscapes, and high inter-trial trajectory variance exhibit high ED. The growth-rate and input-sensitivity findings situate the result within the edge-of-chaos and reservoir-computing literature on critical computation in recurrent networks [9, 16, 10]. Third, I show that architecture (network size) is a probabilistic substrate for this regime: the empirical effect of capacity on ED is statistically robust ($d = 1.08$ between small and large networks) but architecturally modest ($\rho^2 \approx 0.15$), consistent with a probability-landscape rather than a determinative reading of the architecture–dynamics relationship.

2 Embodiment Dependence and Corrected Ghost Conditions

2.1 Embodiment Dependence as a Measurable Variable

Embodiment dependence as I use the term is operationally defined: the magnitude of neural-state divergence when sensorimotor coupling is disrupted by intervention. The term picks out a property of the dynamical relation between a network’s internal state and its sensory input, not a property of the cognitive system as a whole. A network can be highly embodiment-dependent without the body being constitutive of cognition in any of the senses canvassed in the embodied cognition literature [1, 12], and a network might satisfy criteria of constitutive

embodiment while exhibiting low ED if its evolved dynamics maintain stability under sensory perturbation. I make no claim that ED measures constitutive embodiment, sense-making, or any other concept central to enactive theory. ED measures one dimension of the relation between neural and sensorimotor dynamics, and what follows reports how this dimension varies with computational capacity and dynamical regime in evolved CTRNN controllers.

2.2 Interventionism and the Choice of Dependent Variable

The methodology adopted here is interventionist in Woodward’s sense [21]: causal relationships are robust to the extent that they hold across variations in background conditions, and ED is assessed by intervening on the sensorimotor loop and measuring whether neural dynamics survive the intervention. Ghost conditions are interventions that disrupt this loop, and their use in evolutionary robotics has a clear pedigree: Di Paolo [11], Beer [4], and Izquierdo-Torres and Di Paolo [15] all use ghost-condition methods productively, with divergence measured at the level of behavior or task performance. The methodology I propose here extends this tradition to a different dependent variable: neural-state divergence rather than behavioral divergence.

The choice of dependent variable carries a constraint that the existing tradition does not face. When the metric is divergence in neural state, and the intervention is sensory replay to a deterministic network from identical initial states, divergence is zero by construction. The same deterministic update rule applied to the same input from the same initial condition produces the same output. The intervention provides no information about whether the network’s dynamics depend on coupling, only that the simulation is deterministic. The corrected conditions below replace sensory replay with interventions that vary input rather than replay it, making neural-state divergence a usable dependent variable.

The methodological extension is empirically motivated. Across the 60 evolved conditions reported below, behavioral (motor-output) divergence and neural-state divergence correlate within-condition at $\rho \approx 0.6$ – 0.76 . But the capacity–ED relationship (Spearman $\rho = 0.39$, $p = 0.002$; Section 4.2) appears *only* at the neural-state level. Computed on behavioral divergence, the capacity correlation is $\rho = 0.01$ ($p = 0.96$). The architectural signal is in the dynamics, not in observable behavior, which is why a neural-state-divergence metric is needed to detect it.

2.3 Three Corrected Interventions

Three corrected interventions replace sensory replay with conditions that vary input rather than replay it. *Frozen body* (FB) fixes the body’s spatial position at trial start so that sensory input becomes constant despite ongoing motor commands; this disrupts the feedback loop between motor output and sensory change while leaving input non-zero. *Disconnected* (DC) sets sensory input to zero throughout the trial, testing the network’s behavior in sensory isolation.

Counterfactual (CF) substitutes random input uncorrelated with the body’s actual trajectory, disrupting the specific contingency between action and sensory consequence while preserving input variability.

No single condition implements a perfectly surgical intervention on the loop. FB is fat-handed in Woodward’s terminology, simultaneously disrupting motor-to-sensory feedback and the agent’s changing spatial relationship to the environment. DC tests input dependence broadly rather than coupling dependence specifically. I address this through triangulation [20]: convergent results across the three conditions provide stronger evidence for coupling dependence than any single condition alone, and divergent patterns are themselves informative about which aspect of coupling matters. Each condition individually produces a significant capacity–ED correlation (FB $\rho = 0.39$, DC $\rho = 0.37$, CF $\rho = 0.35$, all $p < 0.01$); DC and CF are nearly perfectly correlated with each other ($r = 0.99$) while FB is partially distinct ($r = 0.24$ with DC, $r = 0.16$ with CF), so the composite captures distinct rather than redundant aspects of coupling.

Woodward’s framework assumes relatively stable causal structures, an assumption that does not transfer cleanly to evolutionary systems where each trajectory produces its own path-dependent causal solution. I therefore apply the framework within a single evolved network rather than across populations: the intervention assesses whether *this* network’s dynamics survive disruption, not whether all networks of a given size do. The probabilistic claim about capacity in Section 4 is a claim about the distribution of such within-network results across the population of evolutionary trajectories.

3 Methods

3.1 CTRNN Agents and Phototaxis Task

Continuous-time recurrent neural networks [3] serve as evolved controllers, with dynamics

$$\tau_i \frac{dy_i}{dt} = -y_i + \sum_j w_{ij} \sigma(y_j + \theta_j) + I_i, \quad (1)$$

where y_i is the state of neuron i , τ_i its time constant, w_{ij} the synaptic weight from j to i , θ_j a bias, σ the sigmoid activation, and I_i external sensory input. Integration uses Euler with $dt = 0.01$. Agents are circular bodies (radius 1.0) with bilateral photoreceptors and differential-drive motors in a 50×50 continuous arena; sensors fall off linearly with range 40 units, and maximum agent speed is 3.0 units per behavioral step. The task is phototaxis to a light source whose position varies across trials. Fitness combines time-averaged proximity to the light (50%) and approach score (50%) over four trials per evaluation, with light positions sampled from arena corners during evolution and uniformly around the perimeter during ghost-condition testing. Each trial runs 500 behavioral steps, with each behavioral step comprising 10 neural integration steps.

3.2 Evolution and Experimental Design

Evolution uses the Microbial Genetic Algorithm [13], a minimal evolutionary algorithm well-suited to small CTRNN search spaces [14]; the use of small evolved networks as toy models follows Beer [5, 6]. Population: 50; generations: 5000; Gaussian mutation $\sigma = 0.2$. Genotype encodes time constants (log-scale, range $[0.5, 5.0]$), weights $([-10, 10])$, and biases $([-10, 10])$. Six network sizes ($n = 2, 3, 4, 5, 6, 8$), ten random seeds per size, yielding 60 conditions. Capacity is operationalized as neuron count n ; Section 4.2 reports state entropy as a richer state-space utilization proxy.

3.3 Embodiment Dependence Score

For each ghost condition, the mean L2 distance between embodied and ghost neural-state trajectories is computed across timesteps and trials. The ED score is a weighted combination of the three condition scores, with FB the most diagnostic intervention (it preserves non-zero input while disrupting feedback) and DC the baseline (it tests input dependence broadly):

$$\text{ED} = 0.5 \min(1.0, D_{\text{FB}}) + 0.3 \min(1.0, D_{\text{CF}}) + 0.2 \min(1.0, D_{\text{DC}}). \quad (2)$$

Each divergence is capped at 1.0 before weighting. Headline correlations are stable across cap thresholds in $\{0.5, 1.0, 1.5, 2.0, 5.0, \infty\}$ (capacity-ED $\rho \in [0.36, 0.40]$, growth-rate-ED $\in [0.54, 0.61]$, input-sensitivity-ED $\in [0.56, 0.58]$; supplementary).

3.4 Dynamical, Weight, and Attractor Characterization

Each evolved network was characterized by three families of measures. The dynamical measures, computed over embodied behavior, are participation ratio, state entropy (Shannon entropy of the binned per-dimension state distribution), inter-trial trajectory distance (mean L2 distance between neural trajectories on different light-position trials), perturbation growth rate, fraction amplifying, and maximum Lyapunov exponent (perturbation-based estimation [7]). At the weight level: mean self-connection (w_{ii}) and maximum real eigenvalue. For attractor geometry, fixed points were located by root-finding from 15 random initial conditions at each of six input amplitudes (0.0–1.0), attractors were classified at operating input, input amplitude was scanned to detect bifurcations, and *input sensitivity* was computed as the range of trajectory variance across the six amplitudes.

4 Results

The analyses below comprise 26 statistical tests; the Benjamini-Hochberg procedure controls the false-discovery rate throughout [8], and 23 of the 26 survive at $q < 0.05$. The Kruskal-Wallis test for the capacity-to-ED relationship is the

marginal case ($q = 0.052$) and is reported as such. Confirmatory and exploratory analyses are distinguished: the capacity–ED correlation and the dynamical characterization were pre-specified, while the weight-level and attractor-geometry analyses below were developed to explain residual variance and should be read as hypothesis-generating.

4.1 Dynamical Regime is the Lead Predictor of ED

Three dynamical measures of recurrent activity predict ED at $\rho > 0.5$: perturbation growth rate (the mean rate at which small perturbations amplify through recurrent dynamics; $\rho = 0.58$, $p < 0.0001$), fraction amplifying ($\rho = 0.49$), and state entropy ($\rho = 0.59$). These three are correlated with each other (growth rate $\leftrightarrow$ fraction amplifying $\rho = 0.92$; growth rate $\leftrightarrow$ state entropy $\rho = 0.74$), indexing the same underlying property: how much recurrent dynamics amplify and spread state-space activity during embodied behavior.

A fourth measure, *inter-trial trajectory distance* (the mean L2 distance between neural trajectories on different light-position trials in the embodied condition), predicts ED at $\rho = 0.80$ ($p < 10^{-13}$), the strongest single predictor in the dataset. The inter-trial measure is largely independent of growth rate (partial $\rho = 0.79$) and input sensitivity (partial $\rho = 0.80$), indexing a feature not captured by the others: how much the network’s dynamics differ across distinct sensorimotor histories. Networks whose dynamics are tightly tied to specific sensorimotor trajectories produce high inter-trial distance and, under ghost interventions, high divergence. The four measures jointly describe the dynamical regime developed further in Section 5.

4.2 Capacity is a Probabilistic Substrate for the Coupling-Dependent Regime

The architectural variable here termed capacity (operationalized as neuron count, with state entropy as a richer state-space utilization proxy) is a probabilistic substrate for the dynamical regime above, not its proximate determinant. Neuron count itself correlates with ED at $\rho = 0.39$ ($p = 0.002$, two-tailed), with a 95% bootstrap CI of $[0.14, 0.59]$ that excludes zero. The substantively informative pairwise comparison is between small ($n = 2\text{--}3$) and large ($n = 6\text{--}8$) networks: Mann-Whitney $U = 84.0$, $p < 0.001$, Cohen’s $d = 1.08$, a large effect by conventional benchmarks. The Kruskal-Wallis test across all six sizes is marginally non-significant after FDR correction ($q = 0.052$), reflecting heavy within-group variance. The architecture–ED relationship is therefore best read as a probability-landscape effect: small networks rarely evolve coupling-dependent dynamics, while large networks do so often but not always.

State entropy, a richer state-space utilization measure than neuron count, correlates with ED at $\rho = 0.59$ and with neuron count at $\rho = 0.58$. Its predictive value largely overlaps with growth rate (partial $\rho = 0.24$, $p = 0.07$, controlling for growth rate). Both measures point in the same direction: larger networks more

often realize state-space configurations from which evolution discovers coupling-dependent regimes.

Table 1. Embodiment dependence by network size, all 60 conditions. Means and standard deviations use the weighted ED score in Eq. 2.

n	Mean $\pm$ SD	Median	Range	CV
2	0.30 ± 0.24	0.24	[0.04, 0.69]	80%
3	0.36 ± 0.25	0.28	[0.03, 0.83]	68%
4	0.59 ± 0.31	0.58	[0.03, 1.00]	53%
5	0.48 ± 0.21	0.43	[0.27, 1.00]	44%
6	0.61 ± 0.31	0.55	[0.19, 1.00]	51%
8	0.66 ± 0.27	0.60	[0.32, 1.00]	41%

The progression is roughly monotonic with a single departure at $n = 5$, where the within-size mean falls below $n = 4$ (Table 1). Mean ED rises from 0.30 at $n = 2$ to 0.66 at $n = 8$, and the coefficient of variation falls from 80% to 41% across the same range, with a similar departure at $n = 6$. Excluding the over-performing seed 137 strengthens the correlation to $\rho = 0.43$ ($n = 54$, $p = 0.001$), confirming the relationship is not driven by that seed. Figure 1 shows the distribution.

4.3 Self-Connection Polarity and Spectral Structure at the Weight Level

At the weight level, the sign of the diagonal element w_{ii} is the strongest predictor of ED ($\rho = +0.37$, $p = 0.003$). High-ED networks (score ≥ 0.70 , $n = 15$) have mean self-connection $+0.03 \pm 11.5$; low-ED networks (score < 0.30 , $n = 17$) have mean self-connection -18.0 ± 16.2 (Mann-Whitney $p = 0.003$, Cohen’s $d = +1.28$). Positive self-connections create excitatory feedback in which input-driven activity amplifies through recurrence, and the resulting dynamics depend on continuing input to maintain equilibrium. Negative self-connections damp activity toward fixed-point attractors that are largely insensitive to input variation. Figure 2 plots the relationship.

Positive self-connections produce high maximum real eigenvalues in the weight matrix (self-connection $\leftrightarrow$ max eigenvalue $\rho = +0.52$, $p < 10^{-4}$), placing the network in a regime where small perturbations amplify through recurrent structure. The eigenvalue itself predicts ED at $\rho = +0.35$ ($p = 0.006$). Controlling for the eigenvalue reduces the partial correlation between self-connection and ED from 0.37 to 0.25 (point estimate; 95% bootstrap CI $[-0.02, 0.45]$, 96% of bootstrap resamples positive), a 33% reduction in the correlation. The bootstrap CI marginally includes zero, so the partial-correlation evidence is consistent with partial mediation but does not robustly establish it on this sample. The directional reading, positive self-connection $\rightarrow$ high eigenvalue $\rightarrow$ ED, is the natural interpretation of the data but is not licensed by partial correlation alone.

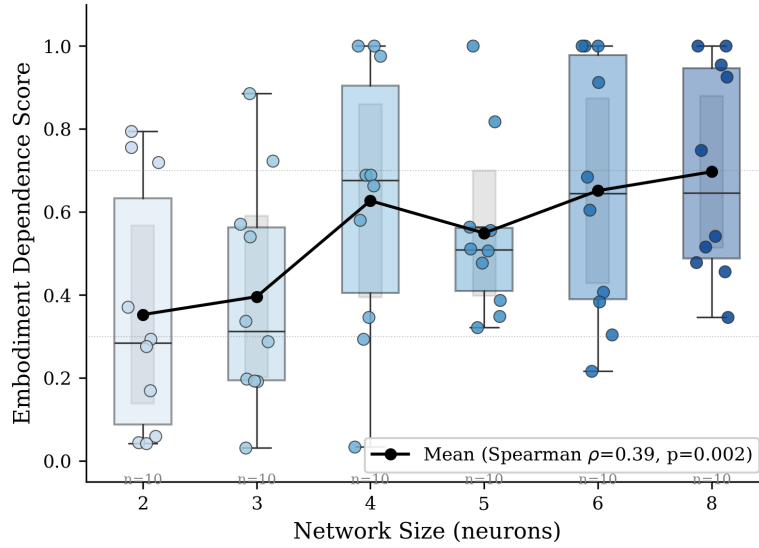


Fig. 1. Embodiment dependence by network size across 60 conditions. Left: scatter with jitter, individual conditions colored by n . Right: distributions per size. Median ED rises and within-size dispersion falls with n .

4.4 Input-Sensitive Attractor Landscapes

The attractor-geometry analysis converges with the weight-level findings. Input sensitivity, the range of trajectory variance across six input amplitudes scanned, is the strongest attractor-geometry predictor of ED ($\rho = +0.55$, $p < 0.0001$). It correlates with bifurcation count ($\rho = +0.68$), and the partial-correlation comparison is asymmetric: input sensitivity retains predictive power after controlling for bifurcation count (partial $\rho = +0.42$, $p < 0.001$), while bifurcation count loses its predictive power after controlling for input sensitivity (partial $\rho = +0.06$, $p = 0.67$). High-ED solutions (≥ 0.70 , $n = 15$) undergo on average 2.7 bifurcations across the input range; low-ED solutions (< 0.30 , $n = 17$) undergo zero (Mann-Whitney $p = 0.002$). Table 2 summarizes the attractor-level correlations and Figure 3 shows the input-sensitivity relationship and the bifurcation-count contrast.

The implication is that ED tracks attractor landscapes that restructure as sensory input varies, continuously across the input range and not only at discrete bifurcation thresholds. Ghost-condition disruption is, in effect, a particularly severe form of input change: when the input drops out, the input-sensitive landscape that the evolved dynamics rely on collapses or shifts, producing the divergence the ED score measures. Two regimes emerge from the analysis. Networks with positive self-connection, marginal spectra, and input-sensitive attractors evolve coupling-dependent dynamics and produce high divergence under ghost interventions; networks with strongly negative self-connection, contracting spec-

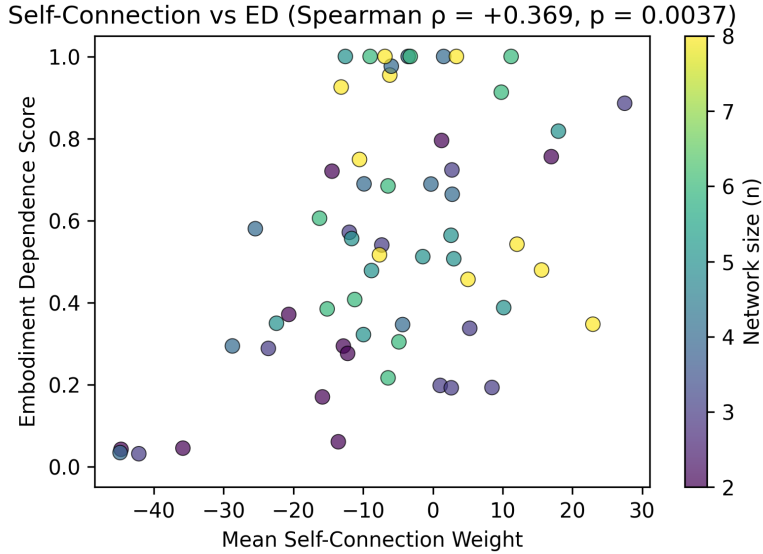


Fig. 2. Self-connection polarity vs. embodiment dependence across 60 conditions ($\rho = +0.37$, $p = 0.003$). High-ED solutions cluster at non-negative self-connections; low-ED solutions cluster at strongly negative self-connections.

Table 2. Attractor-geometry correlations with ED, all 60 conditions.

Metric	Spearman ρ	p
Input sensitivity range	+0.56	< 0.0001
Mean max real eigenvalue (at fixed points)	+0.52	< 0.0001
Trajectory variance at operating input	+0.50	< 0.0001
Number of distinct regimes	+0.44	0.0004
Number of bifurcations	+0.42	0.0007
Stable fixed-point fraction	-0.33	0.011

tra, and input-independent fixed-point attractors evolve coupling-independent dynamics and produce low divergence. The 28 intermediate networks lie along a continuum between these extremes.

5 Discussion

The dynamical signature identified here (positive self-connection, high spectral radius, input-sensitive attractor landscape, high inter-trial trajectory variance) describes networks operating in the edge-of-chaos regime the reservoir-computing literature has identified as the substrate for input-driven recurrent computation [9, 16, 10]. Modern computational neuroscience has identified the same regime in trained recurrent networks performing context-dependent tasks [19,

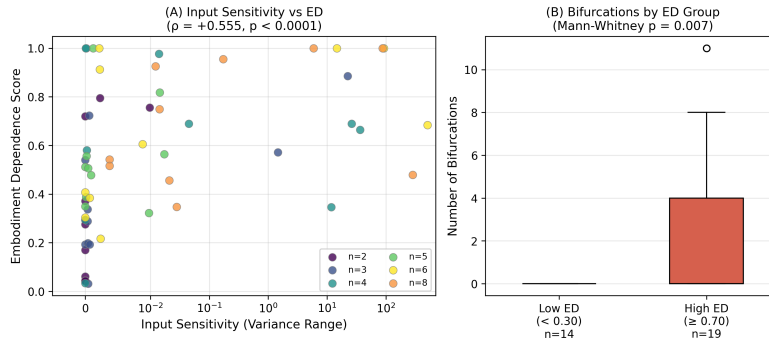


Fig. 3. (A) Input sensitivity vs. embodiment dependence, 60 conditions ($\rho = +0.55$, $p < 0.0001$). (B) Bifurcation count by ED group. High-ED solutions undergo significantly more bifurcations across input conditions (Mann-Whitney $p = 0.002$); low-ED solutions undergo zero.

17]: low-dimensional input-driven structures (line attractors, plane attractors) emerge as the mechanism supporting flexible computation. The convergence between evolved CTRNN controllers and trained RNNs suggests that input-sensitive recurrent dynamics is a general principle for systems whose computation must respond to ongoing sensory structure. Related work on metastability in embodied agents [2] shows extended neural dynamics depend on bidirectional coupling, consistent with the ED finding that coupling-dependent networks collapse under disruption.

The architecture-level claim is modest. Network size explains only about 15% of the variance in ED rank order; growth rate explains about 34%; input sensitivity explains about 31%; inter-trial trajectory distance explains about 63%. (These are squared Spearman rank correlations, not OLS variance-explained quantities; they index rank-order predictive power.) Architecture shifts the probability landscape that evolution samples; whether a given evolutionary run lands on a coupling-dependent or coupling-independent solution is determined by the interaction between architecture and trajectory. Larger networks are more likely to evolve coupling-dependent regimes, but they are not destined to.

Methodologically, the corrected ghost conditions extend the evolutionary-robotics tradition by making neural-state divergence a usable dependent variable. They complement rather than replace the behavioral-divergence ghost conditions of the kind Beer [4], Di Paolo [11], and Izquierdo-Torres and Di Paolo [15] have used productively. The two metrics speak to different questions: behavioral divergence measures task-performance dependence on coupling, while neural-state divergence measures dynamical dependence. The within-condition correlation between the two metrics is moderate ($\rho \approx 0.6$ – 0.76), but the capacity effect appears only at the neural-state level ($\rho = 0.39$ vs. $\rho = 0.01$ behaviorally). For questions about how architecture shapes the coupling-dependent regime, neural-

state divergence is the appropriate metric; for questions about task-performance robustness, the behavioral metric remains the right tool.

Scope. ED measures a property of neural dynamics, not a property of cognition or sense-making. The findings here are compatible with Adams and Aizawa’s [1] view that cognition requires intrinsic representational content, and equally compatible with Di Paolo et al.’s [12] view that cognition emerges from participatory sense-making. ED is a dimension along which evolved controllers vary, not an answer to whether the body is constitutive of cognition. Whether the input-sensitive dynamical regime identified here generalizes to biological recurrent circuits [17] is a question for follow-up work that closes the animal-to-animat loop the present paper does not attempt.

Limitations. (i) Single task (phototaxis); other behaviors may differ. (ii) Single evolutionary algorithm. (iii) Capacity is operationalized as neuron count, with state entropy as a richer proxy. (iv) Weight-level and attractor-level analyses are exploratory; replication is needed. (v) The partial correlation between self-connection and ED controlling for eigenvalue is consistent with partial mediation but its bootstrap CI marginally includes zero; a causal-mediation analysis would be the appropriate follow-up.

6 Conclusion

I have proposed corrected ghost conditions that make neural-state divergence a usable measure of embodiment dependence, applied them across 60 evolved CTRNN controllers, and identified a dynamical signature of the coupling-dependent regime. Networks with high perturbation growth rate, input-sensitive attractor landscapes, and high inter-trial trajectory variance exhibit high ED. Architecture shifts the probability that evolution discovers this regime; it does not determine it.

These contributions matter for three audiences. For evolutionary robotics, the corrected ghost conditions extend interventionist methods from behavioral to neural-state divergence. For the embodied cognition debate, embodiment dependence offers a graded empirical property where the constitutive-versus-causal question has stalled. For computational neuroscience, the dynamical signature converges with the edge-of-chaos regime and with the input-driven attractor structures of trained recurrent networks, suggesting a shared mechanism behind a phenomenon described in three vocabularies.

Disclosures. The author has no competing interests to declare. AI-based tools (Claude, Anthropic) were used for editorial assistance only; all scientific content, design, implementation, analysis, and interpretation are the author’s own. Code, raw and capped divergence values, sensitivity analyses, bootstrap CIs, and the multiple-comparison-correction table are available at <https://github.com/collinlucken/when-does-the-body-matter>.

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